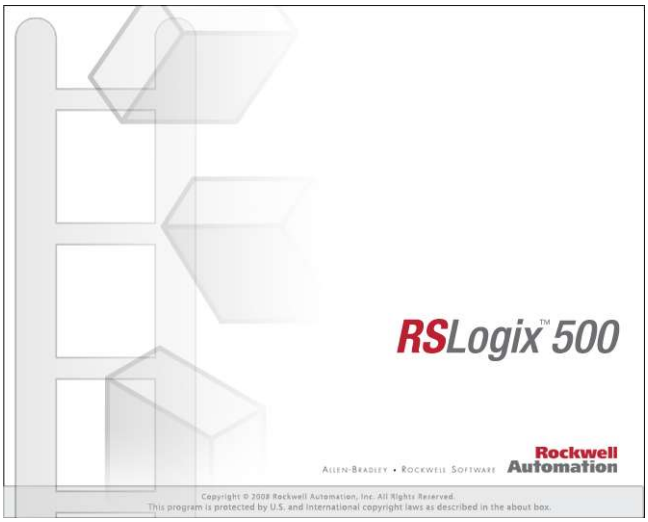
RSLogix Micro Project Report



Processor Information

Processor Type: Bul.1763 MicroLogix 1100 Series B

Processor Name: UNTITLED

Total Memory Used: 813 Instruction Words Used - 129 Data Table Words Used

Total Memory Left: 5843 Instruction Words Left

Program Files: 13

Data Files: 11

Program ID: 78a1

I/O Configuration

0	Bul.1763	MicroLogix 1100 Series B
1		
2		
3		
4		

Channel Configuration

CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Edit Resource/Owner Timeout: 60
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Passthru Link ID: 1
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Write Protected: No
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Comms Servicing Selection: Yes
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex Message Servicing Selection: Yes
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 1st AWA Append Character: \d
CHANNEL 0 (SYSTEM) - Driver: DF1 Full Duplex 2nd AWA Append Character: \a

Source ID: 1 (decimal)
Baud: 9600
Parity: NONE
Control Line : No Handshaking
Error Detection: CRC
Embedded Responses: Auto Detect
Duplicate Packet Detect: Yes
ACK Timeout(x20 ms): 50
NAK Retries: 3
ENQ Retries: 3

CHANNEL 1 (SYSTEM) - Driver: Ethernet
CHANNEL 1 (SYSTEM) - Driver: Ethernet Edit Resource/Owner Timeout: 60
CHANNEL 1 (SYSTEM) - Driver: Ethernet Passthru Link ID: 1
CHANNEL 1 (SYSTEM) - Driver: Ethernet Write Protected: No
CHANNEL 1 (SYSTEM) - Driver: Ethernet Comms Servicing Selection: Yes
CHANNEL 1 (SYSTEM) - Driver: Ethernet Message Servicing Selection: Yes

Hardware Address: 00:00:00:00:00:00
IP Address: 0.0.0.0
Subnet Mask: 0.0.0.0
Gateway Address: 0.0.0.0
Msg Connection Timeout (x 1mS): 15000
Msg Reply Timeout (x mS): 3000
Inactivity Timeout (x Min): 30
Bootp Enable: Yes
Dhcp Enable No
SNMP Enable: No
HTTP Enable: Yes
Auto Negotiate Enable: Yes
Port Speed Enable: 10/100 Mbps Full Duplex/Half Duplex
Contact:
Location:

Program File List

Name	Number	Type	Rungs	Debug	Bytes
[SYSTEM]	0	SYS	0	No	0
	1	SYS	0	No	0
	2	LADDER	12	No	102
IO	3	LADDER	13	No	468
ALARMS	4	LADDER	18	No	576
CTRL	5	LADDER	8	No	325
HOA	6	LADDER	23	No	803
SIM	7	LADDER	7	No	476
MODE	8	LADDER	4	No	112
BACKWASH	9	LADDER	16	No	455
LEVEL	10	LADDER	5	No	113
HOURMETER	11	LADDER	8	No	323
REVIEW	12	LADDER	7	No	466

Data File List

Name	Number	Type	Scope	Debug	Words	Elements	Last
OUTPUT	0	O	Global	No	12	4	O:3
INPUT	1	I	Global	No	18	6	I:5
STATUS	2	S	Global	No	0	66	S:65
BINARY	3	B	Global	No	7	7	B3:6
TIMER	4	T	Global	No	45	15	T4:14
COUNTER	5	C	Global	No	3	1	C5:0
CONTROL	6	R	Global	No	3	1	R6:0
INTEGER	7	N	Global	No	13	13	N7:12
FLOAT	8	F	Global	No	24	12	F8:11
DIGITAL IO	9	B	Global	No	1	1	B9:0
ANALOG IO	10	N	Global	No	3	3	N10:2

Call all ladders in the program.

0000 — IO
JSR
Jump To Subroutine
SBR File Number U:3

0001 — Alarms
JSR
Jump To Subroutine
SBR File Number U:4

0002 — Control Logic
JSR
Jump To Subroutine
SBR File Number U:5

0003 — HOA
JSR
Jump To Subroutine
SBR File Number U:6

0004 — Simulation
JSR
Jump To Subroutine
SBR File Number U:7

0005 — System Mode
JSR
Jump To Subroutine
SBR File Number U:8

0006 — Backwash Cycle
JSR
Jump To Subroutine
SBR File Number U:9

0007 — Level Control
JSR
Jump To Subroutine
SBR File Number U:10

0008 — Hourmeter
JSR
Jump To Subroutine
SBR File Number U:11

0009 — Review / Testing Ladder
JSR
Jump To Subroutine
SBR File Number U:12

Prevent division by zero from faulting processor

0010 — Overflow Trap
S:5
U
0

0011 — END

Map output control bits to output devices. In a REAL program, the addresses on the right would be O: addresses, but this program is written solely for use with Emulate, and therefore no I: or O: address are used. Instead, we're using bit addresses and integer addresses so that the SIM ladder can generate artificial input signals.

0000
Flow Switch FS-1 B9:0 0 — Flow Switch FS-1 Bit FS1 B3:0 0

0001
Solenoid Valve SV-1 Bit SV1 B3:0 1 — Solenoid Valve SV-1 B9:0 1

0002
Solenoid Valve SV-2 Bit SV2 B3:0 2 — Solenoid Valve SV-2 B9:0 2

0003
Solenoid Valve SV-3 Bit SV3 B3:0 3 — Solenoid Valve SV-3 B9:0 3

0004
Solenoid Valve SV-4 Bit SV4 B3:0 4 — Solenoid Valve SV-4 B9:0 4

0005
Solenoid Valve SV-5 Bit SV5 B3:0 5 — Solenoid Valve SV-5 B9:0 5

0006
Solenoid Valve SV-6 Bit SV6 B3:0 6 — Solenoid Valve SV-6 B9:0 6

0007
Pump P-1 Bit P1 B3:0 7 — Pump P-1 B9:0 7

0008

Pressure Transmitter
PT-1 (0-20 psi)
Scaled Value
PT1_SV

SCP
Scale w/Parameters

Input	N10:0
	0<
Input Min.	0.0
	0.0<
Input Max.	16383.0
	16383.0<
Scaled Min.	0.0
	0.0<
Scaled Max.	20.0
	20.0<
Output	F8:0
	0.0<

Pressure Transmitter
PT-1 (0-20 psi) In

LES
Less Than (A<B)

Source A	N10:0
	0<
Source B	0
	0<

Pressure Transmitter
PT-1 (0-20 psi)
Scaled Value
PT1_SV

MOV
Move

Source	0.0
	0.0<
Dest	F8:0
	0.0<

Pressure Transmitter
PT-1 (0-20 psi) In

GRT
Greater Than (A>B)

Source A	N10:0
	0<
Source B	16383
	16383<

Pressure Transmitter
PT-1 (0-20 psi)
Scaled Value
PT1_SV

MOV
Move

Source	20.0
	20.0<
Dest	F8:0
	0.0<

0009

Pressure Transmitter
PT-2 (0-20 psi)
Scaled Value
PT2_SV

SCP
Scale w/Parameters
Input N10:1
0<
Input Min. 0.0
0.0<
Input Max. 16383.0
16383.0<
Scaled Min. 0.0
0.0<
Scaled Max. 20.0
20.0<
Output F8:1
0.0<

Pressure Transmitter
PT-2 (0-20 psi) In

LES
Less Than (A<B)
Source A N10:1
0<
Source B 0
0<

Pressure Transmitter
PT-2 (0-20 psi)
Scaled Value
PT2_SV

MOV
Move
Source 0.0
0.0<
Dest F8:1
0.0<

Pressure Transmitter
PT-2 (0-20 psi) In

GRT
Greater Than (A>B)
Source A N10:1
0<
Source B 16383
16383<

Pressure Transmitter
PT-2 (0-20 psi)
Scaled Value
PT2_SV

MOV
Move
Source 20.0
20.0<
Dest F8:1
0.0<

0010

Preliminary DP

SUB
Subtract
Source A F8:0
0.0<
Source B F8:1
0.0<
Dest F8:11
0.0<

Differential
Pressure
DIFF_PRES

ABS
Absolute Value
Source F8:11
0.0<
Dest F8:3
0.0<

0011

Level Transmitter
LT-1 (0-100 %)
Scaled Value
LT1_SV

SCP
Scale w/Parameters

Input	N10:2
	0<
Input Min.	0.0
	0.0<
Input Max.	16383.0
	16383.0<
Scaled Min.	0.0
	0.0<
Scaled Max.	100.0
	100.0<
Output	F8:2
	0.0<

Level Transmitter
LT-1 (0-100 %) In

LES
Less Than (A<B)

Source A	N10:2
	0<
Source B	0
	0<

Level Transmitter
LT-1 (0-100 %)
Scaled Value
LT1_SV

MOV
Move

Source	0.0
	0.0<
Dest	F8:2
	0.0<

Level Transmitter
LT-1 (0-100 %) In

GRT
Greater Than (A>B)

Source A	N10:2
	0<
Source B	16383
	16383<

Level Transmitter
LT-1 (0-100 %)
Scaled Value
LT1_SV

MOV
Move

Source	20.0
	20.0<
Dest	F8:2
	0.0<

0012

END

High Pressure alarm to prevent explosions and messes

0000

Pressure Transmitter
PT-1 (0-20 psi)
Scaled Value
PT1_SV

GRT
Greater Than (A>B)
Source A F8:0
0.0<
Source B F8:5
15.0<

Alarm Reset
ALARM_RESET
B3:3
9

High Pressure Alarm
Delay

TON
Timer On Delay
Timer T4:1
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

Pressure Transmitter
PT-2 (0-20 psi)
Scaled Value
PT2_SV

GRT
Greater Than (A>B)
Source A F8:1
0.0<
Source B F8:5
15.0<

0001

High Pressure Alarm
Delay
T4:1
DN

ONS
B3:3
ONS
13

High Pressure Alarm
Trigger
B3:3
14

0002

High Pressure Alarm
Trigger
B3:3
14

High Pressure Alarm
Bit
B3:4
7

High Pressure Alarm
Bit
B3:4
7

Alarm Reset
ALARM_RESET
B3:3
9

0003

High Pressure Alarm
Trigger
B3:3
14

High Pressure
Notification Bit
PRES_HH_ALARM
B3:4
8

High Pressure
Notification Bit
PRES_HH_ALARM
B3:4
8

Alarm Reset
ALARM_RESET
B3:3
9

Alarm Silence
ALARM_SILENCE
B3:3
10

Low flow alarm (to protect the pump from cavitation)

0004

Pump P-1 Bit
P1
B3:0
7

Flow Switch
FS-1 Bit
FS1
B3:0
0

Alarm Reset
ALARM_RESET
B3:3
9

Low Flow Alarm Delay

TON
Timer On Delay
Timer T4:2
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

0005
Low Flow Alarm Delay T4:2 DN
ONS B3:3 ONS 15
Low Flow Alarm Trigger B3:4 0

0006
Low Flow Alarm Trigger B3:4 0
Low Flow Alarm Bit B3:4 9
Low Flow Alarm Bit B3:4 9
Alarm Reset ALARM_RESET B3:3 9

0007
Low Flow Alarm Trigger B3:4 0
Low Flow Notification Bit LOW_FLOW_ALARM B3:4 10
Low Flow Notification Bit LOW_FLOW_ALARM B3:4 10
Alarm Reset ALARM_RESET B3:3 9
Alarm Silence ALARM_SILENCE B3:3 10

Low level alarm to prevent sediment at the bottom of the tank from being sent downstream

0008
Level Transmitter LT-1 (0-100 %) Scaled Value LT1_SV
LES
Less Than (A<B)
Source A F8:2 0.0<
Source B F8:6 10.0<
Alarm Reset ALARM_RESET B3:3 9
Storage Tank LL Alarm Delay
TON
Timer On Delay
Timer T4:3
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

0009
Storage Tank LL Alarm Delay T4:3 DN
ONS B3:4 ONS 1
Storage Tank LL Alarm Trigger B3:4 2

0010
Storage Tank LL Alarm Trigger B3:4 2
Storage Tank LL Alarm Bit B3:4 11
Storage Tank LL Alarm Bit B3:4 11
Alarm Reset ALARM_RESET B3:3 9

0011

Storage Tank LL Alarm Trigger B3:4 2

Storage Tank LL Notification Bit LEVEL_LL_ALARM B3:4 12

Storage Tank LL Notification Bit LEVEL_LL_ALARM B3:4 12

Alarm Reset ALARM_RESET B3:3 9

Alarm Silence ALARM_SILENCE B3:3 10

High level alarm to prevent flooding

0012

Level Transmitter LT-1 (0-100 %) Scaled Value LT1_SV

GRT Greater Than (A>B)
Source A F8:2 0.0<
Source B F8:7 90.0<

Alarm Reset ALARM_RESET B3:3 9

Storage Tank HH Alarm Delay

TON Timer On Delay
Timer T4:4
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

0013

Storage Tank HH Alarm Delay T4:4 DN

ONS B3:4 3

Storage Tank HH Alarm Trigger B3:4 4

0014

Storage Tank HH Alarm Trigger B3:4 4

Storage Tank HH Alarm Bit B3:4 13

Storage Tank HH Alarm Bit B3:4 13

Alarm Reset ALARM_RESET B3:3 9

0015

Storage Tank HH Alarm Trigger B3:4 4

Storage Tank HH Notification Bit LEVEL_HH_ALARM B3:4 14

Storage Tank HH Notification Bit LEVEL_HH_ALARM B3:4 14

Alarm Reset ALARM_RESET B3:3 9

Alarm Silence ALARM_SILENCE B3:3 10

Creating a critical alarm bit for use within the logic to determine whether the system can run

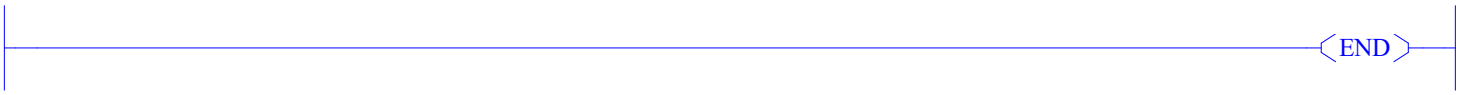
0016

High Pressure Alarm Bit B3:4 7

Critical Alarm Present B3:4 15

Low Flow Alarm Bit B3:4 9

0017



Abstract logic to energize each of the output devices based on whether its HOA status is HAND or AUTO

0000

Solenoid Valve
SV-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV1_STATUS

EQU
Equal
Source A N7:0
0<
Source B 1
1<

System Running
SYS_RUNNING
B3:3
8

Backwash Running
BACKWASH_RUNNING
B3:4
6

Solenoid Valve
SV-1 Bit
SV1
B3:0
1

0001

Solenoid Valve
SV-2 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV2_STATUS

EQU
Equal
Source A N7:1
0<
Source B 1
1<

System Running
SYS_RUNNING
B3:3
8

Backwash Running
BACKWASH_RUNNING
B3:4
6

Solenoid Valve
SV-2 Bit
SV2
B3:0
2

0002

Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS

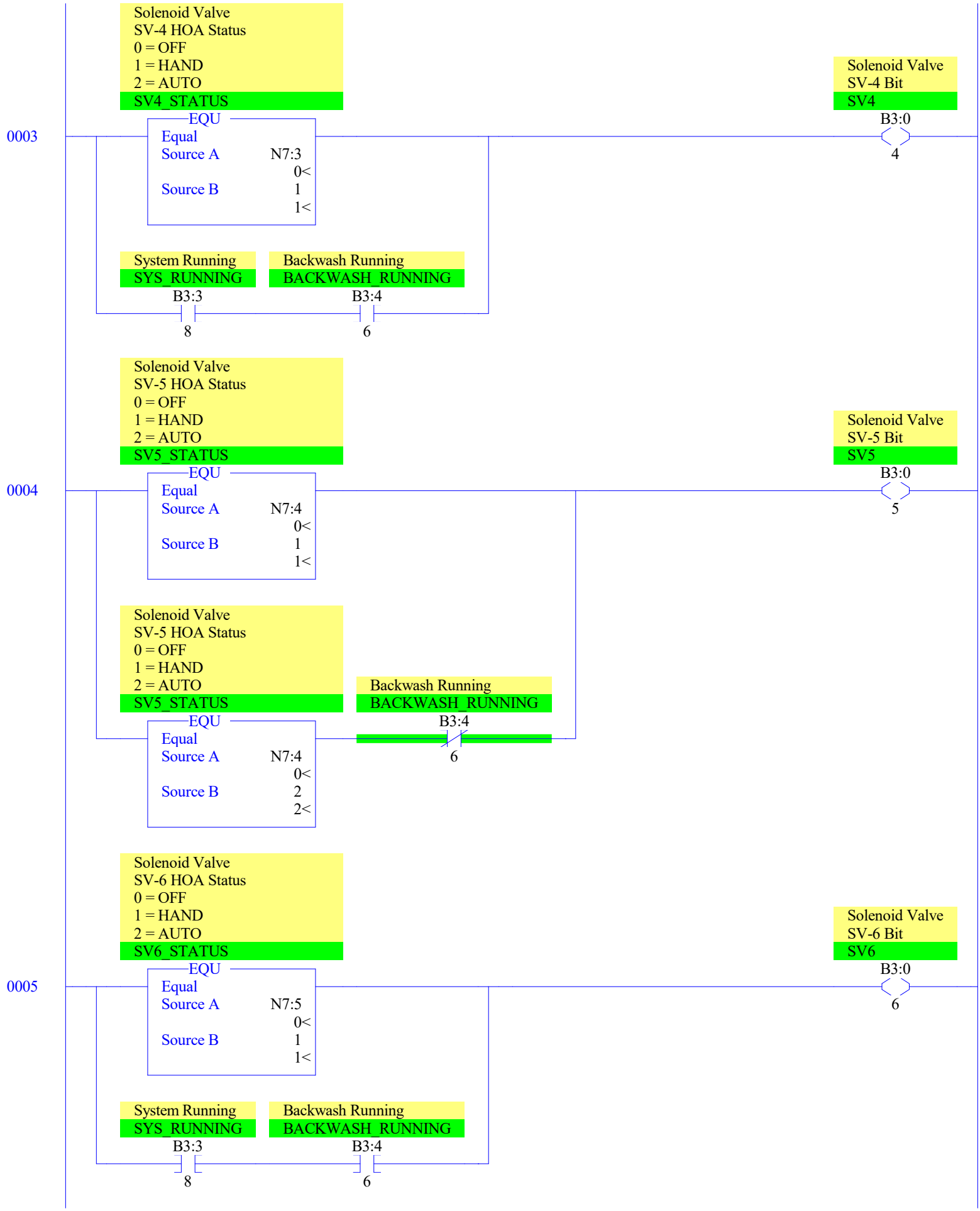
EQU
Equal
Source A N7:2
0<
Source B 1
1<

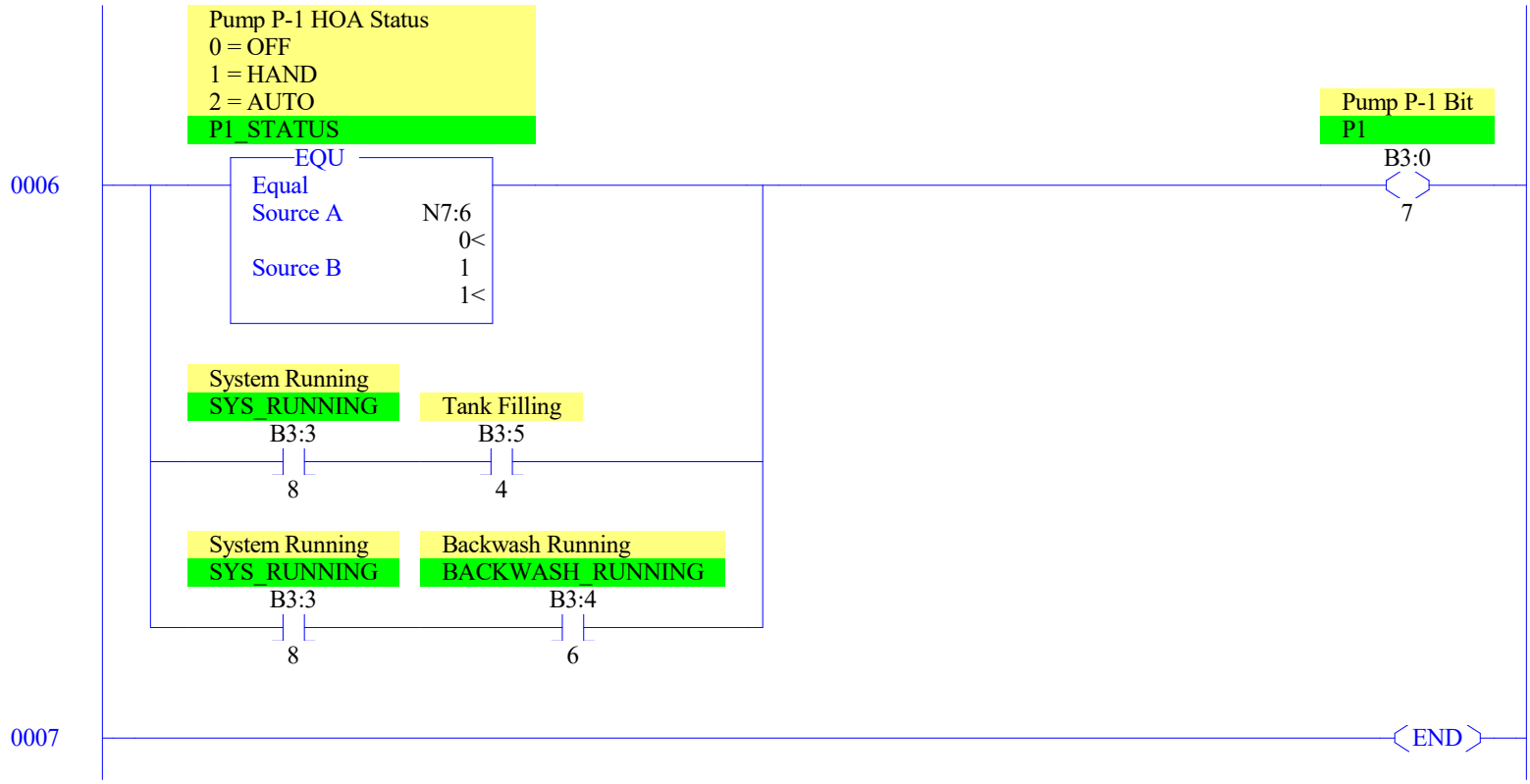
Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS

EQU
Equal
Source A N7:2
0<
Source B 2
2<

Backwash Running
BACKWASH_RUNNING
B3:4
6

Solenoid Valve
SV-3 Bit
SV3
B3:0
3


0003
Solenoid Valve
SV-4 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV4_STATUS
EQU
Equal
Source A N7:3
0<
Source B 1
1<
System Running
SYS_RUNNING
B3:3
8
Backwash Running
BACKWASH_RUNNING
B3:4
6
Solenoid Valve
SV-4 Bit
SV4
B3:0
4
0004
Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS
EQU
Equal
Source A N7:4
0<
Source B 1
1<
Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS
EQU
Equal
Source A N7:4
0<
Source B 2
2<
Backwash Running
BACKWASH_RUNNING
B3:4
6
Solenoid Valve
SV-5 Bit
SV5
B3:0
5
0005
Solenoid Valve
SV-6 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV6_STATUS
EQU
Equal
Source A N7:5
0<
Source B 1
1<
System Running
SYS_RUNNING
B3:3
8
Backwash Running
BACKWASH_RUNNING
B3:4
6
Solenoid Valve
SV-6 Bit
SV6
B3:0
6


Pump P-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
P1_STATUS
EQU
Equal
Source A N7:6
0<
Source B 1
1<
Pump P-1 Bit
P1
B3:0
7
0006
System Running
SYS_RUNNING
B3:3
8
Tank Filling
B3:5
4
System Running
SYS_RUNNING
B3:3
8
Backwash Running
BACKWASH_RUNNING
B3:4
6
0007
END

0000

Solenoid Valve
SV-1 OFF Pushbutton
SV1_OFF_PB
B3:0
8

First Pass
S:1
15

ONS
B3:1
ONS
13

Solenoid Valve
SV-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV1_STATUS

MOV
Move
Source 0
0<
Dest N7:0
0<

0001

Solenoid Valve
SV-1 HAND Pushbutton
SV1_HAND_PB
B3:0
9

ONS
B3:1
ONS
14

Solenoid Valve
SV-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV1_STATUS

MOV
Move
Source 1
1<
Dest N7:0
0<

0002

Solenoid Valve
SV-1 AUTO Pushbutton
SV1_AUTO_PB
B3:0
10

ONS
B3:1
ONS
15

Solenoid Valve
SV-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV1_STATUS

MOV
Move
Source 2
2<
Dest N7:0
0<

0003

Solenoid Valve
SV-2 OFF Pushbutton
SV2_OFF_PB
B3:0
11

First Pass
S:1
15

ONS
B3:2
ONS
0

Solenoid Valve
SV-2 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV2_STATUS

MOV
Move
Source 0
0<
Dest N7:1
0<

0004

Solenoid Valve
SV-2 HAND Pushbutton
SV2_HAND_PB
B3:0
12

ONS
B3:2
ONS
1

Solenoid Valve
SV-2 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV2_STATUS

MOV
Move
Source 1
1<
Dest N7:1
0<

0005

Solenoid Valve
SV-2 AUTO Pushbutton
SV2_AUTO_PB
B3:0
13

ONS
B3:2
ONS
2

Solenoid Valve
SV-2 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV2_STATUS

MOV
Move
Source 2
2<
Dest N7:1
0<

0006

Solenoid Valve
SV-3 OFF Pushbutton
SV3_OFF_PB
B3:0
14

First Pass
S:1
15

ONS
B3:2
ONS
3

Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS

MOV
Move
Source 0
0<
Dest N7:2
0<

0007

Solenoid Valve
SV-3 HAND Pushbutton
SV3_HAND_PB
B3:0
15

ONS
B3:2
ONS
4

Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS

MOV
Move
Source 1
1<
Dest N7:2
0<

0008

Solenoid Valve
SV-3 AUTO Pushbutton
SV3_AUTO_PB
B3:1
0

ONS
B3:2
ONS
5

Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS

MOV
Move
Source 2
2<
Dest N7:2
0<

0009

Solenoid Valve
SV-4 OFF Pushbutton
SV4_OFF_PB
B3:1
1

First Pass
S:1
15

ONS
B3:2
ONS
6

Solenoid Valve
SV-4 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV4_STATUS

MOV
Move
Source 0
0<
Dest N7:3
0<

0010

Solenoid Valve
SV-4 HAND Pushbutton
SV4_HAND_PB
B3:1
2

ONS
B3:2
ONS
7

Solenoid Valve
SV-4 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV4_STATUS

MOV
Move
Source 1
1<
Dest N7:3
0<

0011

Solenoid Valve
SV-4 AUTO Pushbutton
SV4_AUTO_PB
B3:1
3

ONS
B3:2
ONS
8

Solenoid Valve
SV-4 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV4_STATUS

MOV
Move
Source 2
2<
Dest N7:3
0<

0012

Solenoid Valve
SV-5 OFF Pushbutton
SV5_OFF_PB
B3:1
4

First Pass
S:1
15

ONS
B3:2
ONS
9

Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS

MOV
Move
Source 0
0<
Dest N7:4
0<

0013

Solenoid Valve
SV-5 HAND Pushbutton
SV5_HAND_PB
B3:1
5

ONS
B3:2
ONS
10

Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS

MOV
Move
Source 1
1<
Dest N7:4
0<

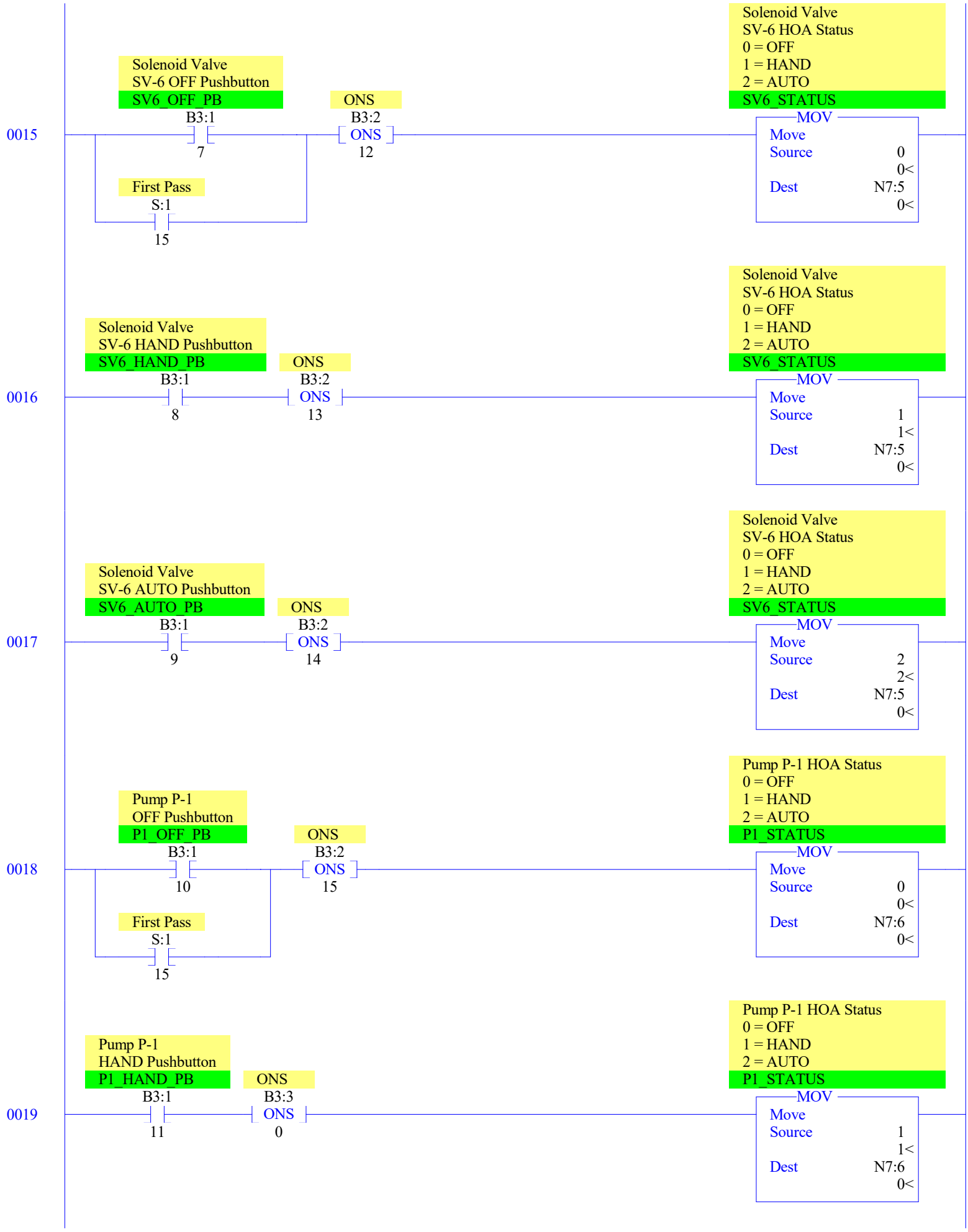
0014

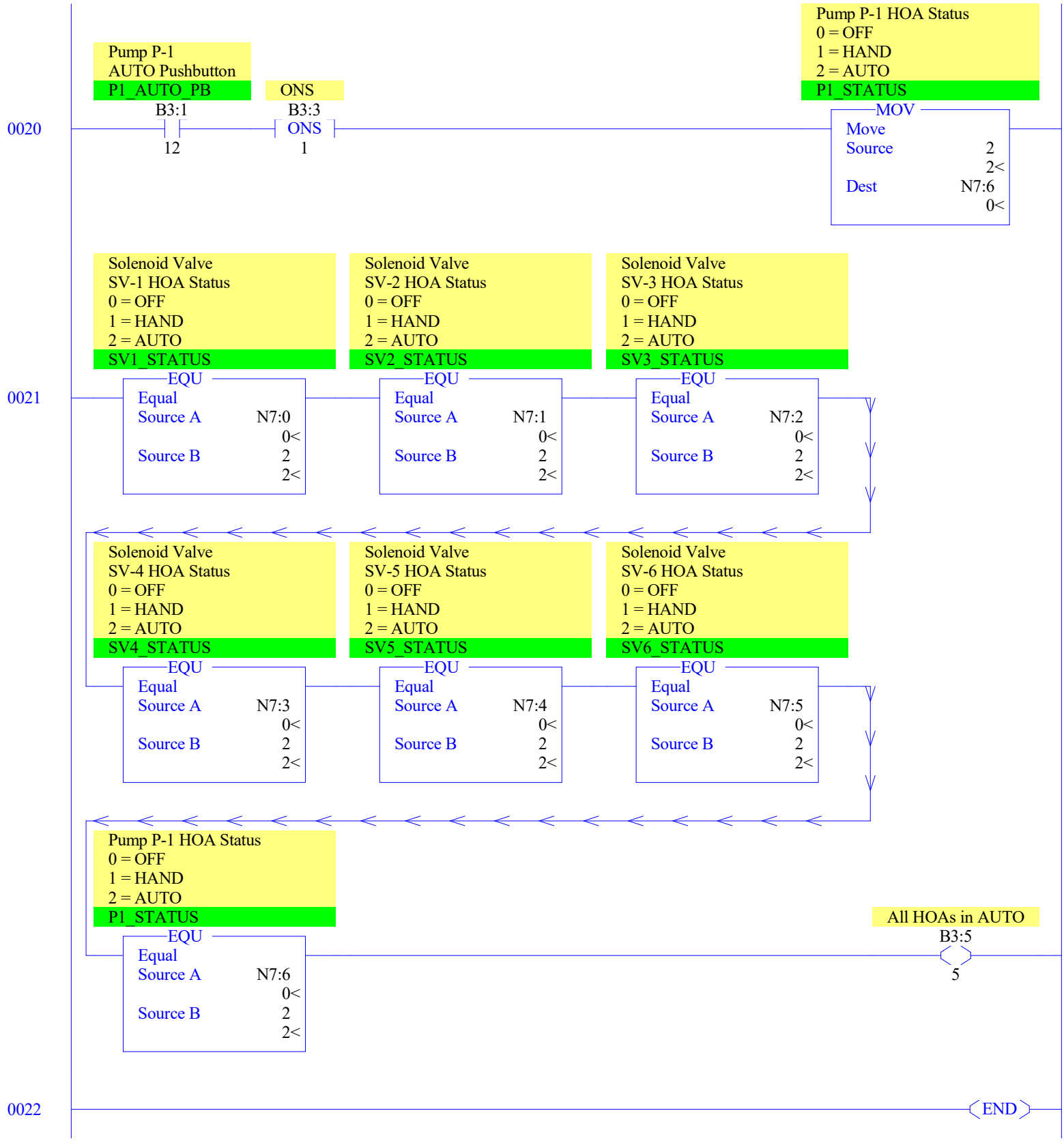
Solenoid Valve
SV-5 AUTO Pushbutton
SV5_AUTO_PB
B3:1
6

ONS
B3:2
ONS
11

Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS

MOV
Move
Source 2
2<
Dest N7:4
0<


0015
Solenoid Valve
SV-6 OFF Pushbutton
SV6_OFF_PB
B3:1
7
First Pass
S:1
15
ONS
B3:2
ONS
12
Solenoid Valve
SV-6 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV6_STATUS
MOV
Move
Source 0
0<
Dest N7:5
0<
0016
Solenoid Valve
SV-6 HAND Pushbutton
SV6_HAND_PB
B3:1
8
ONS
B3:2
ONS
13
Solenoid Valve
SV-6 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV6_STATUS
MOV
Move
Source 1
1<
Dest N7:5
0<
0017
Solenoid Valve
SV-6 AUTO Pushbutton
SV6_AUTO_PB
B3:1
9
ONS
B3:2
ONS
14
Solenoid Valve
SV-6 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV6_STATUS
MOV
Move
Source 2
2<
Dest N7:5
0<
0018
Pump P-1
OFF Pushbutton
P1_OFF_PB
B3:1
10
First Pass
S:1
15
ONS
B3:2
ONS
15
Pump P-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
P1_STATUS
MOV
Move
Source 0
0<
Dest N7:6
0<
0019
Pump P-1
HAND Pushbutton
P1_HAND_PB
B3:1
11
ONS
B3:3
ONS
0
Pump P-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
P1_STATUS
MOV
Move
Source 1
1<
Dest N7:6
0<


0020
Pump P-1
AUTO Pushbutton
P1_AUTO_PB
B3:1
12
ONS
B3:3
ONS
1
Pump P-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
P1_STATUS
MOV
Move
Source 2
2<
Dest N7:6
0<
0021
Solenoid Valve
SV-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV1_STATUS
EQU
Equal
Source A N7:0
0<
Source B 2
2<
Solenoid Valve
SV-2 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV2_STATUS
EQU
Equal
Source A N7:1
0<
Source B 2
2<
Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS
EQU
Equal
Source A N7:2
0<
Source B 2
2<
Solenoid Valve
SV-4 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV4_STATUS
EQU
Equal
Source A N7:3
0<
Source B 2
2<
Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS
EQU
Equal
Source A N7:4
0<
Source B 2
2<
Solenoid Valve
SV-6 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV6_STATUS
EQU
Equal
Source A N7:5
0<
Source B 2
2<
Pump P-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
P1_STATUS
EQU
Equal
Source A N7:6
0<
Source B 2
2<
All HOAs in AUTO
B3:5
5
0022
END

0000

Simulation logic governs the input signals allowing the system logic to function as if it were running in a real-world, dynamic environment.

Simulation
Calculation Timer

T4:9
DN

Simulation
Calculation Timer

TON
Timer On Delay
Timer T4:9
Time Base 0.001
Preset 500<
Accum 0<

EN
DN

0001

Simulation Calculation Timer
T4:9
DN

Pump P-1 Bit
P1
B3:0
7

Backwash Running
BACKWASH_RUNNING
B3:4
6

Level Transmitter
LT-1 (0-100 %) In

ADD
Add
Source A N10:2
0<
Source B 80
80<
Dest N10:2
0<

Filter Resistence

ADD
Add
Source A F8:10
0.0<
Source B 6.0
6.0<
Dest F8:10
0.0<

Pump P-1 Bit
P1
B3:0
7

Backwash Running
BACKWASH_RUNNING
B3:4
6

Level Transmitter
LT-1 (0-100 %) In

SUB
Subtract
Source A N10:2
0<
Source B 12
12<
Dest N10:2
0<

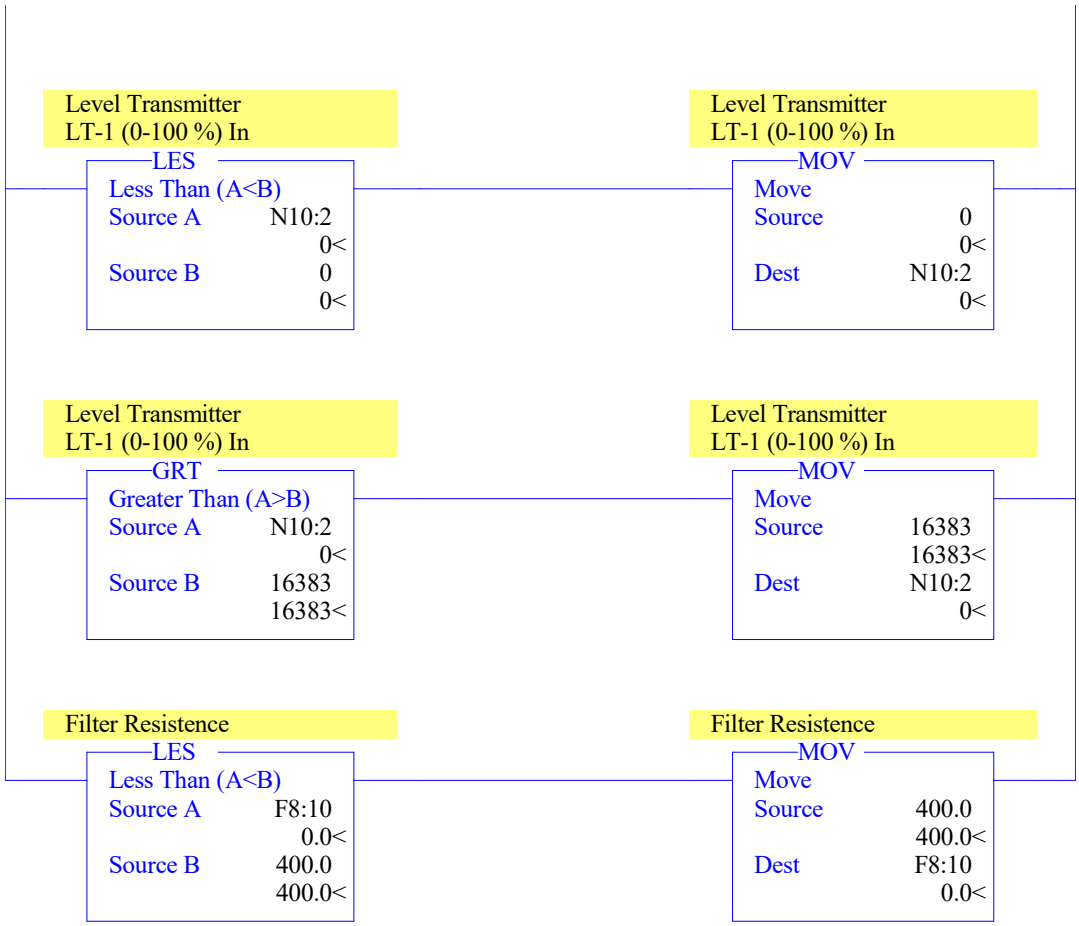
Filter Resistence

SUB
Subtract
Source A F8:10
0.0<
Source B 22.0
22.0<
Dest F8:10
0.0<

System Running
SYS_RUNNING
B3:3
8

Level Transmitter
LT-1 (0-100 %) In

SUB
Subtract
Source A N10:2
0<
Source B 17
17<
Dest N10:2
0<


Level Transmitter
LT-1 (0-100 %) In
LES
Less Than (A<B)
Source A N10:2
0<
Source B 0
0<
Level Transmitter
LT-1 (0-100 %) In
MOV
Move
Source 0
0<
Dest N10:2
0<
Level Transmitter
LT-1 (0-100 %) In
GRT
Greater Than (A>B)
Source A N10:2
0<
Source B 16383
16383<
Level Transmitter
LT-1 (0-100 %) In
MOV
Move
Source 16383
16383<
Dest N10:2
0<
Filter Resistence
LES
Less Than (A<B)
Source A F8:10
0.0<
Source B 400.0
400.0<
Filter Resistence
MOV
Move
Source 400.0
400.0<
Dest F8:10
0.0<

0002

Pump P-1 Bit
P1
B3:0
7

Flow Switch Close Delay

TON
Timer On Delay
Timer T4:10
Time Base 1.0
Preset 2<
Accum 0<
EN
DN

Backwash Running
BACKWASH_RUNNING
B3:4
6

Pressure Transmitter
PT-2 (0-20 psi) In

MOV
Move
Source 1000
1000<
Dest N10:1
0<

Pressure Transmitter
PT-1 (0-20 psi) In

ADD
Add
Source A N10:1
0<
Source B F8:10
0.0<
Dest N10:0
0<

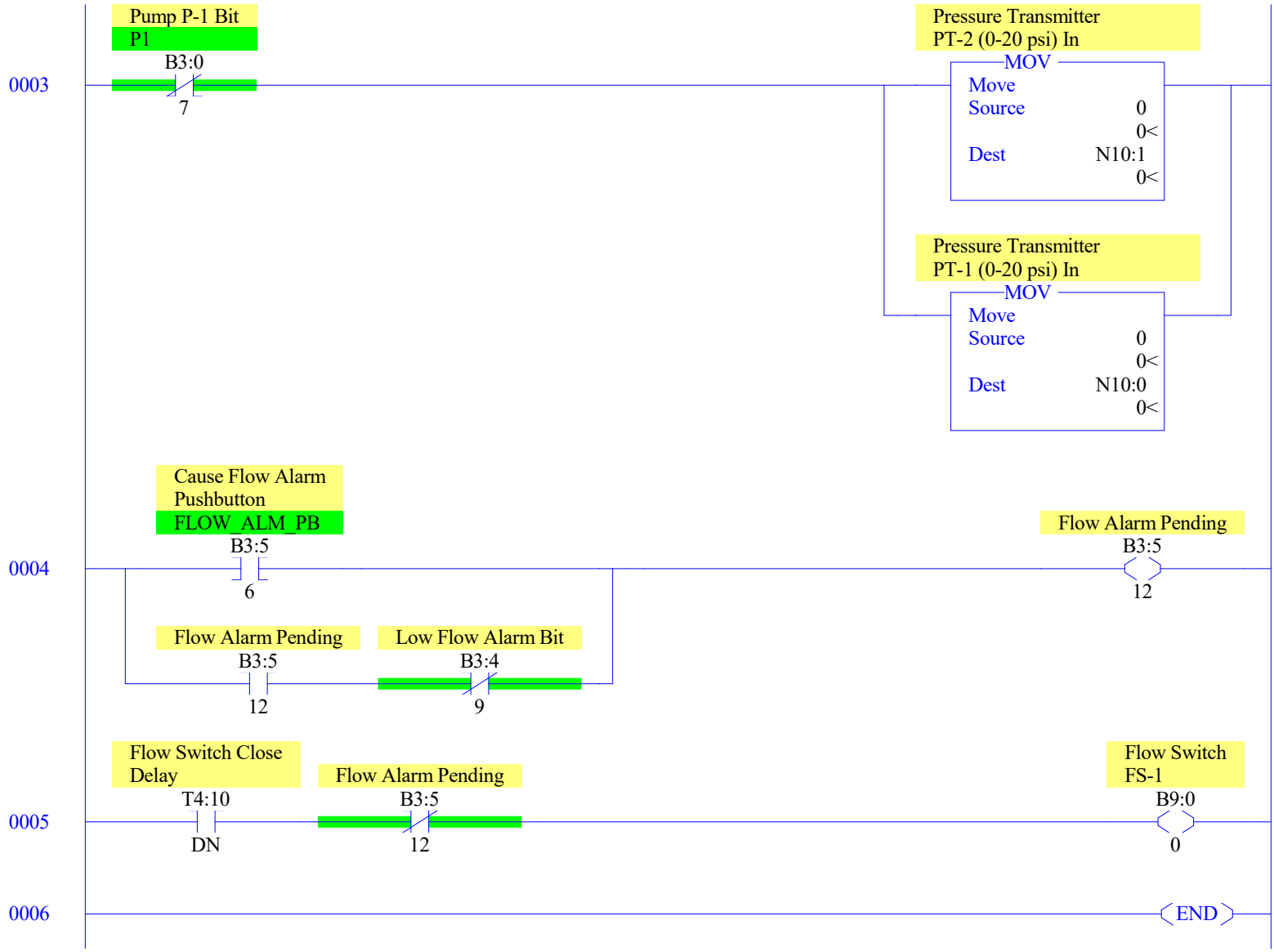
Backwash Running
BACKWASH_RUNNING
B3:4
6

Pressure Transmitter
PT-1 (0-20 psi) In

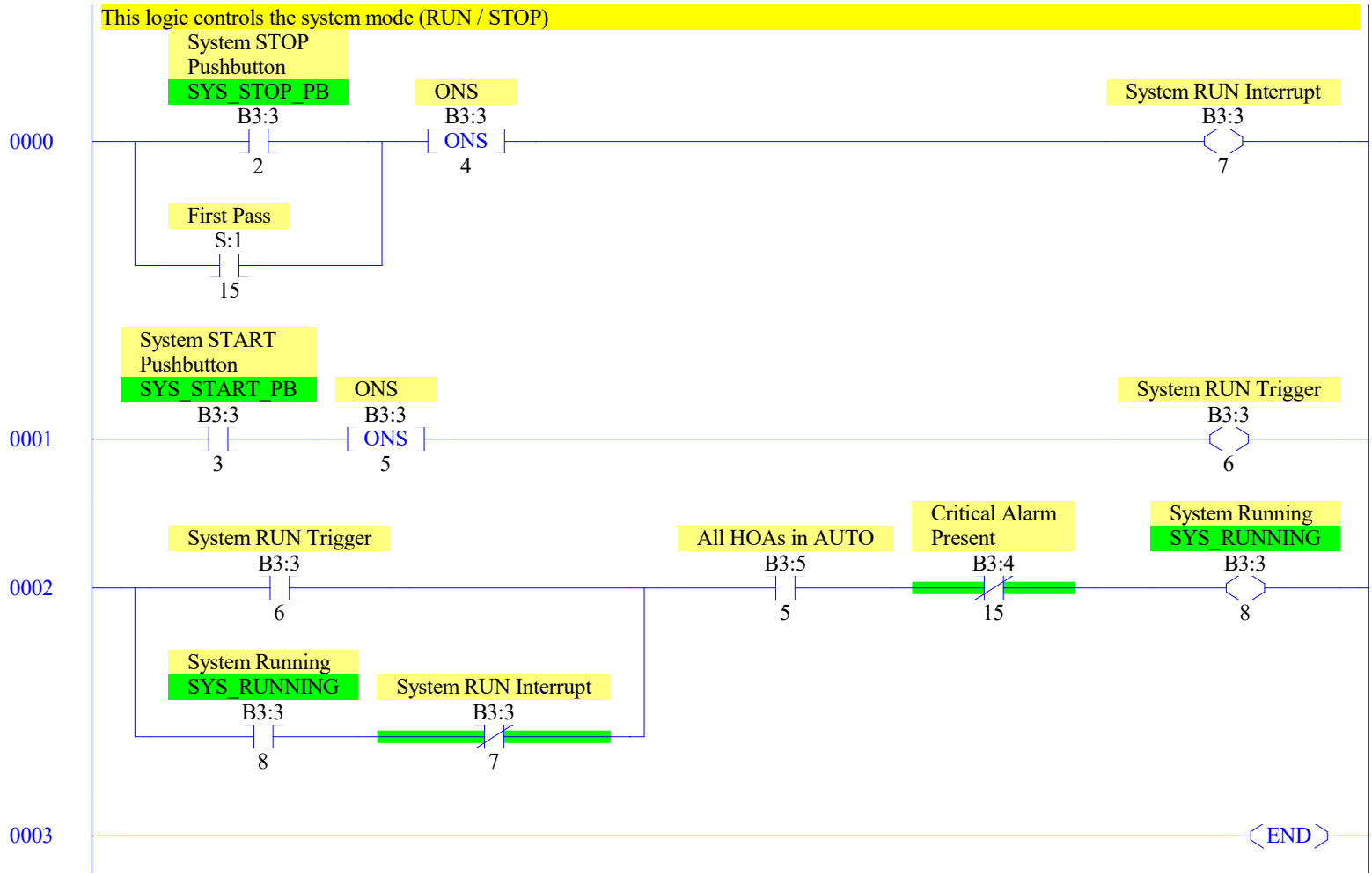
MOV
Move
Source 700
700<
Dest N10:0
0<

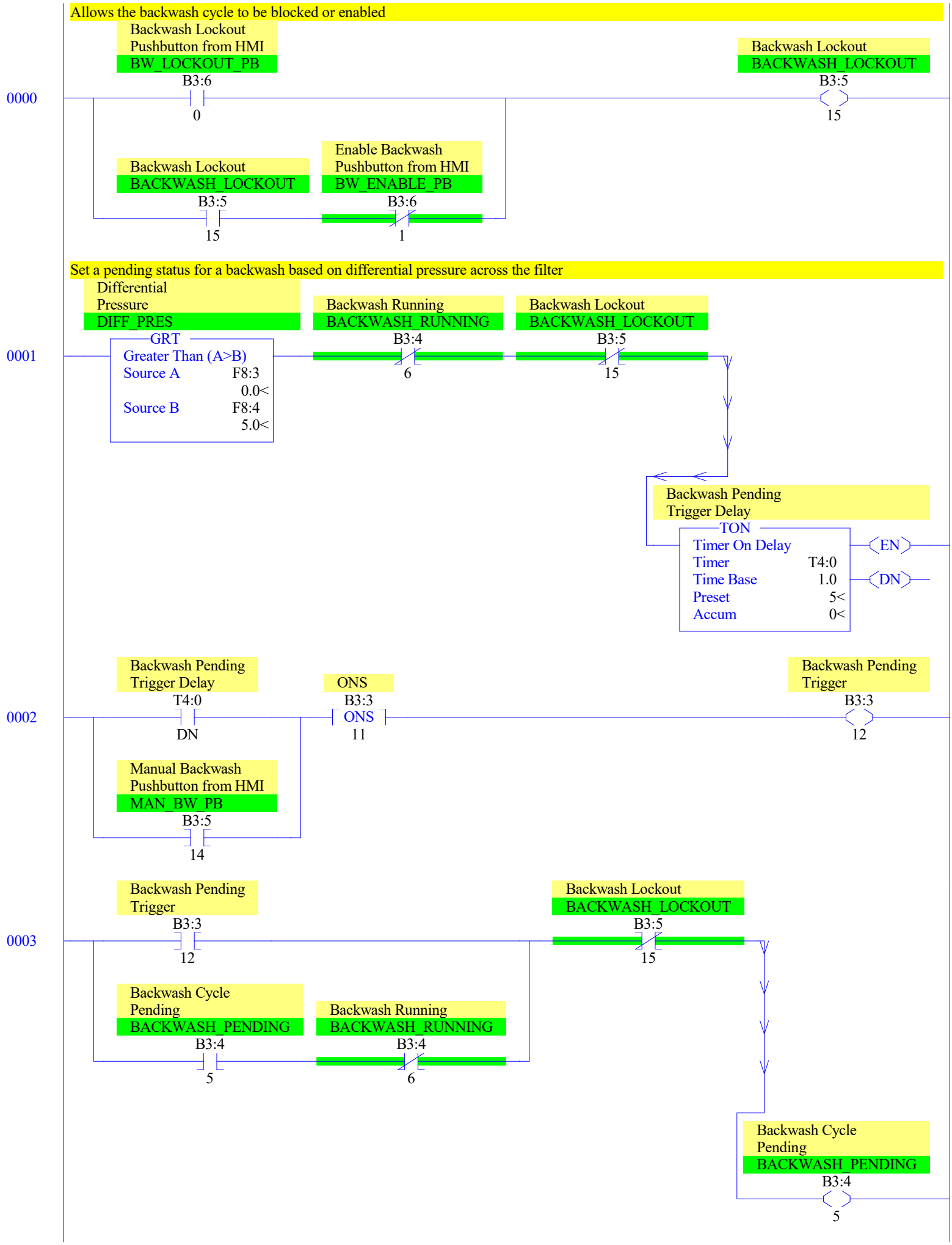
Pressure Transmitter
PT-2 (0-20 psi) In

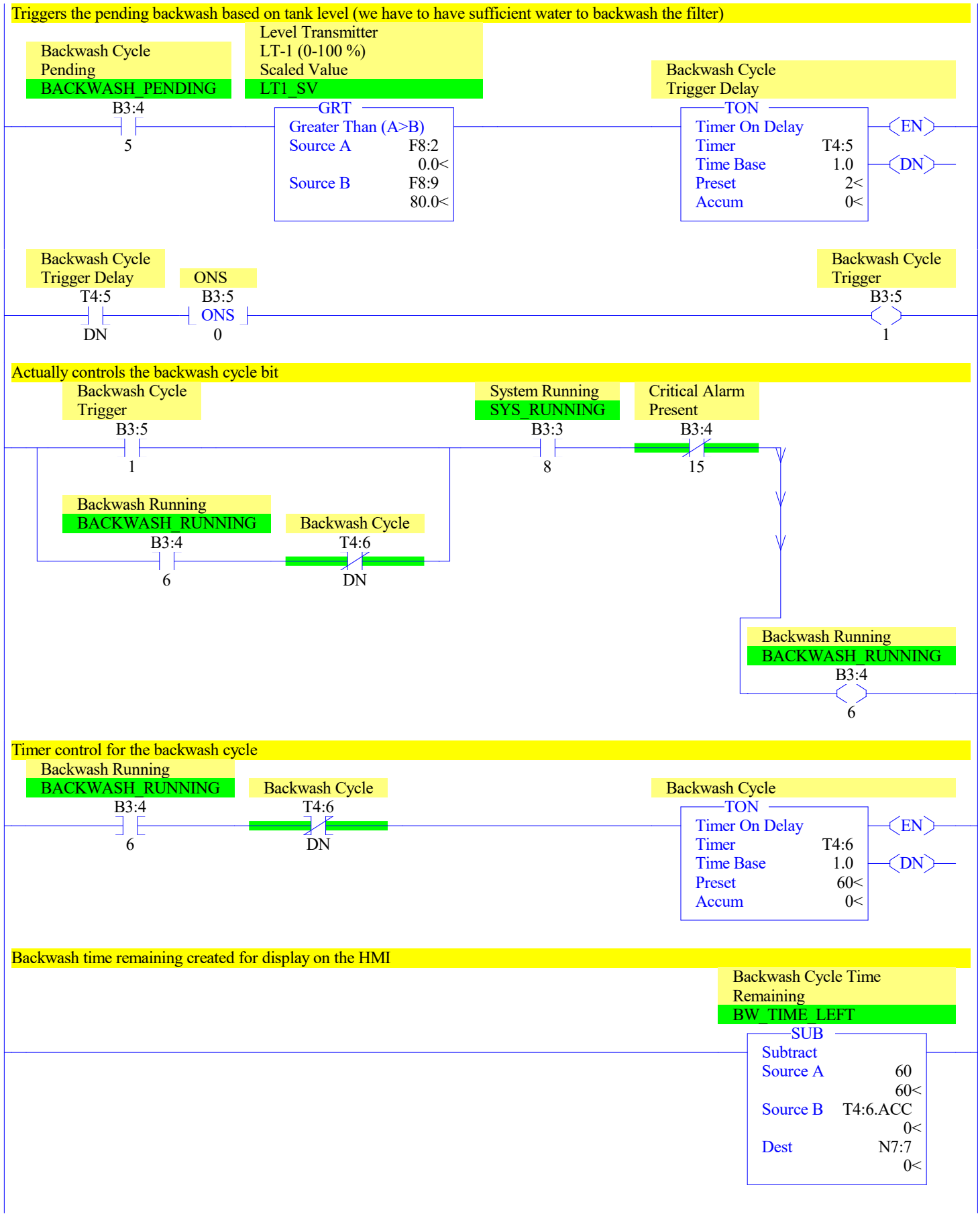
ADD
Add
Source A N10:0
0<
Source B F8:10
0.0<
Dest N10:1
0<


0003
Pump P-1 Bit
P1
B3:0
7
Pressure Transmitter
PT-2 (0-20 psi) In
MOV
Move
Source 0
0<
Dest N10:1
0<
Pressure Transmitter
PT-1 (0-20 psi) In
MOV
Move
Source 0
0<
Dest N10:0
0<
0004
Cause Flow Alarm
Pushbutton
FLOW_ALM_PB
B3:5
6
Flow Alarm Pending
B3:5
12
Flow Alarm Pending
B3:5
12
Low Flow Alarm Bit
B3:4
9
0005
Flow Switch Close
Delay
T4:10
DN
Flow Alarm Pending
B3:5
12
Flow Switch
FS-1
B9:0
0
0006
END

LAD 8 - MODE --- Total Rungs in File = 4




Allows the backwash cycle to be blocked or enabled
0000
Backwash Lockout Pushbutton from HMI
BW_LOCKOUT_PB
B3:6
0
Backwash Lockout
BACKWASH_LOCKOUT
B3:5
15
Backwash Lockout
BACKWASH_LOCKOUT
B3:5
15
Enable Backwash Pushbutton from HMI
BW_ENABLE_PB
B3:6
1
Set a pending status for a backwash based on differential pressure across the filter
0001
Differential Pressure
DIFF_PRES
GRT
Greater Than (A>B)
Source A F8:3
0.0<
Source B F8:4
5.0<
Backwash Running
BACKWASH_RUNNING
B3:4
6
Backwash Lockout
BACKWASH_LOCKOUT
B3:5
15
Backwash Pending Trigger Delay
TON
Timer On Delay
Timer T4:0
Time Base 1.0
Preset 5<
Accum 0<
EN
DN
0002
Backwash Pending Trigger Delay
T4:0
DN
ONS
B3:3
ONS
11
Backwash Pending Trigger
B3:3
12
Manual Backwash Pushbutton from HMI
MAN_BW_PB
B3:5
14
0003
Backwash Pending Trigger
B3:3
12
Backwash Lockout
BACKWASH_LOCKOUT
B3:5
15
Backwash Cycle Pending
BACKWASH_PENDING
B3:4
5
Backwash Running
BACKWASH_RUNNING
B3:4
6
Backwash Cycle Pending
BACKWASH_PENDING
B3:4
5


Triggers the pending backwash based on tank level (we have to have sufficient water to backwash the filter)
0004
Backwash Cycle Pending
BACKWASH_PENDING
B3:4
5
Level Transmitter
LT-1 (0-100 %)
Scaled Value
LT1_SV
GRT
Greater Than (A>B)
Source A F8:2
0.0<
Source B F8:9
80.0<
Backwash Cycle Trigger Delay
TON
Timer On Delay
Timer T4:5
Time Base 1.0
Preset 2<
Accum 0<
EN
DN
0005
Backwash Cycle Trigger Delay
T4:5
DN
ONS
B3:5
ONS
0
Backwash Cycle Trigger
B3:5
1
Actually controls the backwash cycle bit
0006
Backwash Cycle Trigger
B3:5
1
System Running
SYS_RUNNING
B3:3
8
Critical Alarm Present
B3:4
15
Backwash Running
BACKWASH_RUNNING
B3:4
6
Backwash Cycle
T4:6
DN
Backwash Running
BACKWASH_RUNNING
B3:4
6
Timer control for the backwash cycle
0007
Backwash Running
BACKWASH_RUNNING
B3:4
6
Backwash Cycle
T4:6
DN
Backwash Cycle
TON
Timer On Delay
Timer T4:6
Time Base 1.0
Preset 60<
Accum 0<
EN
DN
Backwash time remaining created for display on the HMI
0008
Backwash Cycle Time Remaining
BW_TIME_LEFT
SUB
Subtract
Source A 60
60<
Source B T4:6.ACC
0<
Dest N7:7
0<

Cycle counter for backwash

0009

Backwash Running
BACKWASH_RUNNING
B3:4
6

ONS
B3:6
ONS
2

Backwash Cycle Count
BACKWASH_COUNT

ADD
Add
Source A N7:8
0<
Source B 1
1<
Dest N7:8
0<

Data log trigger for BACKWASH table in the SQL database

0010

Backwash Running
BACKWASH_RUNNING
B3:4
6

ONS
B3:6
ONS
5

Trigger Backwash Data Log
B3:6
6

0011

Trigger Backwash Data Log
B3:6
6

Backwash Data Log Pulse Width
T4:12
EN

Backwash Data Log Pulse Width
T4:12
DN

Backwash Data Log Pulse Width

TON
Timer On Delay
Timer T4:12
Time Base 0.001
Preset 500<
Accum 0<
EN
DN

0012

Backwash Data Log Pulse Width
T4:12
EN

Log Backwash Data Bit
LOG_BW_DATA
B3:6
7

Reset action for the backwash cycle count

0013

Reset Backwash Cycle Count Pushbutton from HMI
RESET_BW_COUNT_PB
B3:6
3

Backwash Cycle Count Reset Delay

TON
Timer On Delay
Timer T4:11
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

0014

Backwash Cycle Count Reset Delay
T4:11
DN

ONS
B3:6
ONS
4

Backwash Cycle Count
BACKWASH_COUNT

MOV
Move
Source 0
0<
Dest N7:8
0<

0015

END

Fills and drains the tank by controlling the pump based on level

0000

Level Transmitter
LT-1 (0-100 %)
Scaled Value
LT1_SV

LES
Less Than (A<B)
Source A F8:2
0.0<
Source B F8:8
20.0<

System Running
SYS_RUNNING
B3:3
8

Fill Start Delay

TON
Timer On Delay
Timer T4:7
Time Base 1.0
Preset 2<
Accum 0<

EN
DN

0001

Fill Start Delay
T4:7
DN

ONS
B3:5
ONS
2

Fill Start Trigger
B3:5
3

0002

Level Transmitter
LT-1 (0-100 %)
Scaled Value
LT1_SV

GRT
Greater Than (A>B)
Source A F8:2
0.0<
Source B F8:9
80.0<

Fill Stop Delay

TON
Timer On Delay
Timer T4:8
Time Base 1.0
Preset 2<
Accum 0<

EN
DN

0003

Fill Start Trigger
B3:5
3

Tank Filling
B3:5
4

Tank Filling
B3:5
4

Fill Stop Delay
T4:8
DN

0004

END

Determines WHEN to count hours

0000

System Running
SYS_RUNNING
B3:3
8

Backwash Running
BACKWASH_RUNNING
B3:4
6

Log Time
B3:6
8

Counts by seconds and updates minutes, hours and thousands of hours automatically

0001

Log Time
B3:6
8

Hourmeter
RTO
Retentive Timer On
Timer T4:13
Time Base 1.0
Preset 60<
Accum 0<
EN
DN

Hourmeter Minutes
HOURMETER_SECS
MOV
Move
Source T4:13.ACC
0<
Dest N7:9
0<

0002

Hourmeter
T4:13
DN

Hourmeter Minutes
HOURMETER_MINS
ADD
Add
Source A N7:10
0<
Source B 1
1<
Dest N7:10
0<

Hourmeter
T4:13
RES

0003

Hourmeter Minutes
HOURMETER_MINS
GEQ
Grtr Than or Eql (A>=B)
Source A N7:10
0<
Source B 60
60<

Hourmeter Minutes
HOURMETER_MINS
SUB
Subtract
Source A N7:10
0<
Source B 60
60<
Dest N7:10
0<

Hourmeter Hours
HOURMETER_HRS
ADD
Add
Source A N7:11
0<
Source B 1
1<
Dest N7:11
0<

0004

Hourmeter Hours
HOURMETER_HRS

GEQ
Grtr Than or Eql (A>=B)
Source A N7:11
0<
Source B 1000
1000<

Hourmeter Hours
HOURMETER_HRS

SUB
Subtract
Source A N7:11
0<
Source B 1000
1000<
Dest N7:11
0<

Hourmeter Hours
(x1000)
HOURMETER_KHRS

ADD
Add
Source A N7:12
0<
Source B 1
1<
Dest N7:12
0<

Reset action for the hourmeter

0005

Reset Hourmeter
Pushbutton from HMI
RESET_HOURMETER_PB
B3:6
9

Hourmeter Reset
Delay

TON
Timer On Delay
Timer T4:14
Time Base 1.0
Preset 5<
Accum 0<
EN
DN

0006

Hourmeter Reset Delay
T4:14
DN

ONS
B3:6
ONS
10

Hourmeter Minutes
HOURMETER_SECS
MOV
Move
Source 0
0<
Dest N7:9
0<

Hourmeter Minutes
HOURMETER_MINS
MOV
Move
Source 0
0<
Dest N7:10
0<

Hourmeter Hours
HOURMETER_HRS
MOV
Move
Source 0
0<
Dest N7:11
0<

Hourmeter Hours (x1000)
HOURMETER_KHRS
MOV
Move
Source 0
0<
Dest N7:12
0<

Hourmeter
T4:13
RES

0007

END

This ladder doesn't do ANYTHING. It just makes various system registers avaialable in one place for the programmer (you) to control the system without an HMI and without having to jump back and forth between several ladders

0000

System START Pushbutton
SYS_START_PB
B3:3
3

System STOP Pushbutton
SYS_STOP_PB
B3:3
2

System Running
SYS_RUNNING
B3:3
8

B3:5
7

0001

Alarm Silence
ALARM_SILENCE
B3:3
10

Alarm Reset
ALARM_RESET
B3:3
9

Low Flow Alarm Bit
B3:4
9

High Pressure Alarm Bit
B3:4
7

Storage Tank LL Alarm Bit
B3:4
11

Storage Tank HH Alarm Bit
B3:4
13

Level Transmitter
LT-1 (0-100 %)
Scaled Value
LT1_SV

EQU
Equal
Source A F8:2
0.0<
Source B 9999.0
9999.0<

B3:5
8

0002

Flow Switch FS-1
B9:0
0

Solenoid Valve SV-1 Bit
SV1
B3:0
1

Solenoid Valve SV-2 Bit
SV2
B3:0
2

Solenoid Valve SV-3 Bit
SV3
B3:0
3

Solenoid Valve SV-4 Bit
SV4
B3:0
4

Solenoid Valve SV-5 Bit
SV5
B3:0
5

Solenoid Valve SV-6 Bit
SV6
B3:0
6

Pump P-1 Bit
P1
B3:0
7

B3:5
9

0003

Pressure Transmitter
PT-1 (0-20 psi)
Scaled Value
PT1_SV

EQU
Equal
Source A F8:0
0.0<
Source B 9999.0
9999.0<

Pressure Transmitter
PT-2 (0-20 psi)
Scaled Value
PT2_SV

EQU
Equal
Source A F8:1
0.0<
Source B 9999.0
9999.0<

Filter Resistence

EQU
Equal
Source A F8:10
0.0<
Source B 9999.0
9999.0<

Differential
Pressure
DIFF_PRES

GRT
Greater Than (A>B)
Source A F8:3
0.0<
Source B F8:4
5.0<

Backwash Cycle Time
Remaining
BW_TIME_LEFT

EQU
Equal
Source A N7:7
0<
Source B 9999
9999<

Backwash Cycle
Pending
BACKWASH_PENDING

B3:4
5

Backwash Running
BACKWASH_RUNNING

B3:4
6

B3:5
10

0004

Solenoid Valve
SV-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV1_STATUS

EQU
Equal
Source A N7:0
0<
Source B 2
2<

Solenoid Valve
SV-2 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV2_STATUS

EQU
Equal
Source A N7:1
0<
Source B 2
2<

Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS

EQU
Equal
Source A N7:2
0<
Source B 2
2<

Solenoid Valve
SV-4 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV4_STATUS

EQU
Equal
Source A N7:3
0<
Source B 2
2<

Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS

EQU
Equal
Source A N7:4
0<
Source B 2
2<

Solenoid Valve
SV-6 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV6_STATUS

EQU
Equal
Source A N7:5
0<
Source B 2
2<

Pump P-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
P1_STATUS

EQU
Equal
Source A N7:6
0<
Source B 2
2<

B3:5
11

0005

Put All Devices Into AUTO

B3:5
13

Solenoid Valve
SV-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV1_STATUS

MOV
Move
Source 2
2<
Dest N7:0
0<

Solenoid Valve
SV-2 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV2_STATUS

MOV
Move
Source 2
2<
Dest N7:1
0<

Solenoid Valve
SV-3 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV3_STATUS

MOV
Move
Source 2
2<
Dest N7:2
0<

Solenoid Valve
SV-4 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV4_STATUS

MOV
Move
Source 2
2<
Dest N7:3
0<

Solenoid Valve
SV-5 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV5_STATUS

MOV
Move
Source 2
2<
Dest N7:4
0<

Solenoid Valve
SV-6 HOA Status
0 = OFF
1 = HAND
2 = AUTO
SV6_STATUS

MOV
Move
Source 2
2<
Dest N7:5
0<

Pump P-1 HOA Status
0 = OFF
1 = HAND
2 = AUTO
P1_STATUS

MOV
Move
Source 2
2<
Dest N7:6
0<

0006 END

Data File O0 (bin) -- OUTPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
O:0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
O:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
O:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
O:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B

Data File I1 (bin) -- INPUT

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
I:0.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
I:0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
I:0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
I:0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B
I:0.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B-Anal
I:0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Bul.1763	MicroLogix 1100 Series B-Anal

Main

Processor Mode S:1/0 - S:1/4 = Remote Run
On Power up Go To Run (Mode Behavior) S:1/12 = 0
First Pass S:1/15 = No
Free Running Clock S:4 = 0000-0000-0000-0000

Proc

OS Catalog Number S:57 = 1100
OS Series S:58 = A
OS FRS S:59 =
Processor Catalog Number S:60 =
Processor Series S:61 = A
Processor FRN S:62 =
User Program Type S:63 = 8001h
Compiler Revision Number S:64 =

Scan Times

Maximum (x10 ms) S:22 = 0
Watchdog (x10 ms) S:3 (high byte) = 10
Last 100 uSec Scan Time S:35 = 0
Scan Toggle Bit S:33/9 = 0

Math

Math Overflow Selected S:2/14 = 0
Overflow Trap S:5/0 = 0
Carry S:0/0 = 0
Overflow S:0/1 = 0
Zero Bit S:0/2 = 0
Sign Bit S:0/3 = 0
Math Register (lo word) S:13 = 0
Math Register (high word) S:14-S:13 = 0
Math Register (32 Bit) S:14-S:13 = 0

Chan 0

Processor Mode S:1/0- S:1/4 = Remote Run
Node Address S:15 (low byte) = 0
Baud Rate S:15 (high byte) = ?
Channel Mode S:33/3 = 0
Comms Active S:33/4 = 0
Incoming Cmd Pending S:33/0 = 0
Msg Reply Pending S:33/1 = 0
Outgoing Msg Cmd Pending S:33/2 = 0

Debug

Suspend Code S:7 = 0
Suspend File S:8 = 0

Errors

Fault Override At Power Up S:1/8 = 0
Startup Protection Fault S:1/9 = 0
Major Error Halt S:1/13 = 0
Overflow Trap S:5/0 = 0
Control Register Error S:5/2 = 0
Major Error Executing User Fault Rtn. S:5/3 = 0
Battery Low S:5/11 = 0
Input Filter Selection Modified S:5/13 = 0
ASCII String Manipulation error S:5/15 = 0
Fault Routine S:29 = 0
Major Error S:6 = 0h
Error Description:

Protection

Deny Future Access S:1/14 = No
Data File Overwrite Protection Lost S:36/10 = False

Mem Module

Memory Module Loaded On Boot S:5/8 = 0
Password Mismatch S:5/9 = 0
Load Memory Module On Memory Error S:1/10 = 0
Load Memory Module Always S:1/11 = 0
On Power up Go To Run (Mode Behavior) S:1/12 = 0
Program Compare S:2/9 = 0
Data File Overwrite Protection Lost S:36/10 = 0

Forces

Forces Enabled S:1/5 = Yes
Forces Installed S:1/6 = No

Data File B3 (bin) -- BINARY

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol)	Description
B3:0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
B3:1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
B3:2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
B3:3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
B3:4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
B3:5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
B3:6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

Data File T4 -- TIMER

Offset	EN	TT	DN	BASE	PRE	ACC	(Symbol) Description
T4:0	0	0	0	1.0 sec	5	0	Backwash Pending Trigger Delay
T4:1	0	0	0	1.0 sec	5	0	High Pressure Alarm Delay
T4:2	0	0	0	1.0 sec	5	0	Low Flow Alarm Delay
T4:3	0	0	0	1.0 sec	5	0	Storage Tank LL Alarm Delay
T4:4	0	0	0	1.0 sec	5	0	Storage Tank HH Alarm Delay
T4:5	0	0	0	1.0 sec	2	0	Backwash Cycle Trigger Delay
T4:6	0	0	0	1.0 sec	60	0	Backwash Cycle
T4:7	0	0	0	1.0 sec	2	0	Fill Start Delay
T4:8	0	0	0	1.0 sec	2	0	Fill Stop Delay
T4:9	0	0	0	.001 sec	500	0	Simulation Calculation Timer
T4:10	0	0	0	1.0 sec	2	0	Flow Switch Close Delay
T4:11	0	0	0	1.0 sec	5	0	Backwash Cycle Count Reset Delay
T4:12	0	0	0	.001 sec	500	0	Backwash Data Log Pulse Width
T4:13	0	0	0	1.0 sec	60	0	Hourmeter
T4:14	0	0	0	1.0 sec	5	0	Hourmeter Reset Delay

Data File C5 -- COUNTER

Offset	CU	CD	DN	OV	UN	UA	PRE	ACC	(Symbol)	Description
C5:0	0	0	0	0	0	0	0	0		

Data File R6 -- CONTROL

Offset	EN	EU	DN	EM	ER	UL	IN	FD	LEN	POS	(Symbol)	Description
R6:0	0	0	0	0	0	0	0	0	0	0		

Data File N7 (dec) -- INTEGER

Offset	0	1	2	3	4	5	6	7	8	9
N7:0	0	0	0	0	0	0	0	0	0	0
N7:10	0	0	0							

Data File F8 -- FLOAT

Offset	0	1	2	3	4
F8:0	0	0	0	0	5
F8:5	15	10	90	20	80
F8:10	0	0			

Data File B9 (bin) -- DIGITAL IO

Offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	(Symbol) Description
B9:0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Data File N10 (dec) -- ANALOG IO

Offset	0	1	2	3	4	5	6	7	8	9
N10:0	0	0	0							

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Co
B3:0/0	FS1	Global	Flow Switch FS-1 Bit		
B3:0/1	SV1	Global	Solenoid Valve SV-1 Bit		
B3:0/2	SV2	Global	Solenoid Valve SV-2 Bit		
B3:0/3	SV3	Global	Solenoid Valve SV-3 Bit		
B3:0/4	SV4	Global	Solenoid Valve SV-4 Bit		
B3:0/5	SV5	Global	Solenoid Valve SV-5 Bit		
B3:0/6	SV6	Global	Solenoid Valve SV-6 Bit		
B3:0/7	P1	Global	Pump P-1 Bit		
B3:0/8	SV1_OFF_PB	Global	Solenoid Valve SV-1 OFF Pushbutton		
B3:0/9	SV1_HAND_PB	Global	Solenoid Valve SV-1 HAND Pushbutton		
B3:0/10	SV1_AUTO_PB	Global	Solenoid Valve SV-1 AUTO Pushbutton		
B3:0/11	SV2_OFF_PB	Global	Solenoid Valve SV-2 OFF Pushbutton		
B3:0/12	SV2_HAND_PB	Global	Solenoid Valve SV-2 HAND Pushbutton		
B3:0/13	SV2_AUTO_PB	Global	Solenoid Valve SV-2 AUTO Pushbutton		
B3:0/14	SV3_OFF_PB	Global	Solenoid Valve SV-3 OFF Pushbutton		
B3:0/15	SV3_HAND_PB	Global	Solenoid Valve SV-3 HAND Pushbutton		
B3:1/0	SV3_AUTO_PB	Global	Solenoid Valve SV-3 AUTO Pushbutton		
B3:1/1	SV4_OFF_PB	Global	Solenoid Valve SV-4 OFF Pushbutton		
B3:1/2	SV4_HAND_PB	Global	Solenoid Valve SV-4 HAND Pushbutton		
B3:1/3	SV4_AUTO_PB	Global	Solenoid Valve SV-4 AUTO Pushbutton		
B3:1/4	SV5_OFF_PB	Global	Solenoid Valve SV-5 OFF Pushbutton		
B3:1/5	SV5_HAND_PB	Global	Solenoid Valve SV-5 HAND Pushbutton		
B3:1/6	SV5_AUTO_PB	Global	Solenoid Valve SV-5 AUTO Pushbutton		
B3:1/7	SV6_OFF_PB	Global	Solenoid Valve SV-6 OFF Pushbutton		
B3:1/8	SV6_HAND_PB	Global	Solenoid Valve SV-6 HAND Pushbutton		
B3:1/9	SV6_AUTO_PB	Global	Solenoid Valve SV-6 AUTO Pushbutton		
B3:1/10	P1_OFF_PB	Global	Pump P-1 OFF Pushbutton		
B3:1/11	P1_HAND_PB	Global	Pump P-1 HAND Pushbutton		
B3:1/12	P1_AUTO_PB	Global	Pump P-1 AUTO Pushbutton		
B3:1/13			ONS		
B3:1/14			ONS		
B3:1/15			ONS		
B3:2/0			ONS		
B3:2/1			ONS		
B3:2/2			ONS		
B3:2/3			ONS		
B3:2/4			ONS		
B3:2/5			ONS		
B3:2/6			ONS		
B3:2/7			ONS		
B3:2/8			ONS		
B3:2/9			ONS		
B3:2/10			ONS		
B3:2/11			ONS		
B3:2/12			ONS		
B3:2/13			ONS		
B3:2/14			ONS		
B3:2/15			ONS		
B3:3/0			ONS		
B3:3/1			ONS		
B3:3/2	SYS_STOP_PB	Global	System STOP Pushbutton		
B3:3/3	SYS_START_PB	Global	System START Pushbutton		
B3:3/4			ONS		
B3:3/5			ONS		
B3:3/6			System RUN Trigger		
B3:3/7			System RUN Interrupt		
B3:3/8	SYS_RUNNING	Global	System Running		
B3:3/9	ALARM_RESET	Global	Alarm Reset		
B3:3/10	ALARM_SILENCE	Global	Alarm Silence		
B3:3/11			ONS		
B3:3/12			Backwash Pending Trigger		
B3:3/13			ONS		
B3:3/14			High Pressure Alarm Trigger		
B3:3/15			ONS		
B3:4/0			Low Flow Alarm Trigger		
B3:4/1			ONS		
B3:4/2			Storage Tank LL Alarm Trigger		
B3:4/3			ONS		
B3:4/4			Storage Tank HH Alarm Trigger		
B3:4/5	BACKWASH_PENDING	Global	Backwash Cycle Pending		
B3:4/6	BACKWASH_RUNNING	Global	Backwash Running		
B3:4/7			High Pressure Alarm Bit		
B3:4/8	PRES_HH_ALARM	Global	High Pressure Notification Bit		
B3:4/9			Low Flow Alarm Bit		
B3:4/10	LOW_FLOW_ALARM	Global	Low Flow Notification Bit		
B3:4/11			Storage Tank LL Alarm Bit		
B3:4/12	LEVEL_LL_ALARM	Global	Storage Tank LL Notification Bit		
B3:4/13			Storage Tank HH Alarm Bit		
B3:4/14	LEVEL_HH_ALARM	Global	Storage Tank HH Notification Bit		
B3:4/15			Critical Alarm Present		
B3:5/0			ONS		
B3:5/1			Backwash Cycle Trigger		
B3:5/2			ONS		
B3:5/3			Fill Start Trigger		
B3:5/4			Tank Filling		
B3:5/5			All HOAs in AUTO		
B3:5/6	FLOW_ALM_PB	Global	Cause Flow Alarm Pushbutton		
B3:5/7					
B3:5/8					
B3:5/9					

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Co
B3:5/10					
B3:5/11					
B3:5/12			Flow Alarm Pending		
B3:5/13			Put All Devices Into AUTO		
B3:5/14	MAN_BW_PB	Global	Manual Backwash Pushbutton from HMI		
B3:5/15	BACKWASH_LOCKOUT	Global	Backwash Lockout		
B3:6/0	BW_LOCKOUT_PB	Global	Backwash Lockout Pushbutton from HMI		
B3:6/1	BW_ENABLE_PB	Global	Enable Backwash Pushbutton from HMI		
B3:6/2			ONS		
B3:6/3	RESET_BW_COUNT_PB	Global	Reset Backwash Cycle Count Pushbutton from HMI		
B3:6/4			ONS		
B3:6/5			ONS		
B3:6/6			Trigger Backwash Data Log		
B3:6/7	LOG_BW_DATA	Global	Log Backwash Data Bit		
B3:6/8			Log Time		
B3:6/9	RESET_HOURMETER_PB	Global	Reset Hourmeter Pushbutton from HMI		
B3:6/10			ONS		
B9:0/0			Flow Switch FS-1		
B9:0/1			Solenoid Valve SV-1		
B9:0/2			Solenoid Valve SV-2		
B9:0/3			Solenoid Valve SV-3		
B9:0/4			Solenoid Valve SV-4		
B9:0/5			Solenoid Valve SV-5		
B9:0/6			Solenoid Valve SV-6		
B9:0/7			Pump P-1		
C5:0					
F8:0	PT1_SV	Global	Pressure Transmitter PT-1 (0-20 psi) Scaled Value		
F8:1	PT2_SV	Global	Pressure Transmitter PT-2 (0-20 psi) Scaled Value		
F8:2	LT1_SV	Global	Level Transmitter LT-1 (0-100 %) Scaled Value		
F8:3	DIFF_PRES	Global	Differential Pressure		
F8:4	BACKWASH_SP	Global	Backwash Cycle Trigger Setpoint		
F8:5	PRESSURE_HH_SP	Global	Pressure HH Alarm Setpoint		
F8:6	LEVEL_LL_SP	Global	Storage Tank LL Alarm Setpoint		
F8:7	LEVEL_HH_SP	Global	Storage Tank HH Alarm Setpoint		
F8:8	LEVEL_L_SP	Global	Storage Tank Low Level Setpoint		
F8:9	LEVEL_H_SP	Global	Storage Tank High Level Setpoint		
F8:10			Filter Resistence		
F8:11			Preliminary DP		
N7:0	SV1_STATUS	Global	Solenoid Valve SV-1 HOA Status 0 = OFF 1 = HAND 2 = AUTO		
N7:1	SV2_STATUS	Global	Solenoid Valve SV-2 HOA Status 0 = OFF 1 = HAND 2 = AUTO		
N7:2	SV3_STATUS	Global	Solenoid Valve SV-3 HOA Status 0 = OFF 1 = HAND 2 = AUTO		
N7:3	SV4_STATUS	Global	Solenoid Valve SV-4 HOA Status 0 = OFF 1 = HAND 2 = AUTO		
N7:4	SV5_STATUS	Global	Solenoid Valve SV-5 HOA Status 0 = OFF 1 = HAND 2 = AUTO		
N7:5	SV6_STATUS	Global	Solenoid Valve SV-6 HOA Status 0 = OFF 1 = HAND 2 = AUTO		
N7:6	P1_STATUS	Global	Pump P-1 HOA Status 0 = OFF 1 = HAND 2 = AUTO		
N7:7	BW_TIME_LEFT	Global	Backwash Cycle Time Remaining		
N7:8	BACKWASH_COUNT	Global	Backwash Cycle Count		
N7:9	HOURMETER_SECS	Global	Hourmeter Minutes		
N7:10	HOURMETER_MINS	Global	Hourmeter Minutes		
N7:11	HOURMETER_HRS	Global	Hourmeter Hours		
N7:12	HOURMETER_KHRS	Global	Hourmeter Hours (x1000)		
N10:0			Pressure Transmitter PT-1 (0-20 psi) In		
N10:1			Pressure Transmitter PT-2 (0-20 psi) In		
N10:2			Level Transmitter LT-1 (0-100 %) In		
S:0			Arithmetic Flags		
S:0/0			Processor Arithmetic Carry Flag		
S:0/1			Processor Arithmetic Underflow/ Overflow Flag		
S:0/2			Processor Arithmetic Zero Flag		
S:0/3			Processor Arithmetic Sign Flag		
S:1			Processor Mode Status/ Control		
S:1/0			Processor Mode Bit 0		
S:1/1			Processor Mode Bit 1		
S:1/2			Processor Mode Bit 2		
S:1/3			Processor Mode Bit 3		
S:1/4			Processor Mode Bit 4		
S:1/5			Forces Enabled		
S:1/6			Forces Present		
S:1/7			Comms Active		
S:1/8			Fault Override at Powerup		
S:1/9			Startup Protection Fault		
S:1/10			Load Memory Module on Memory Error		
S:1/11			Load Memory Module Always		
S:1/12			Load Memory Module and RUN		
S:1/13			Major Error Halted		
S:1/14			Access Denied		
S:1/15			First Pass		
S:2/0			STI Pending		
S:2/1			STI Enabled		
S:2/2			STI Executing		
S:2/3			Index Addressing File Range		
S:2/4			Saved with Debug Single Step		
S:2/5			DH-485 Incoming Command Pending		
S:2/6			DH-485 Message Reply Pending		
S:2/7			DH-485 Outgoing Message Command Pending		
S:2/15			Comms Servicing Selection		
S:3			Current Scan Time/ Watchdog Scan Time		
S:4			Time Base		
S:5/0			Overflow Trap		
S:5/2			Control Register Error		
S:5/3			Major Err Detected Executing UserFault Routine		

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Co
S:5/4			M0-M1 Referenced on Disabled Slot		
S:5/8			Memory Module Boot		
S:5/9			Memory Module Password Mismatch		
S:5/10			STI Overflow		
S:5/11			Battery Low		
S:6			Major Error Fault Code		
S:7			Suspend Code		
S:8			Suspend File		
S:9			Active Nodes		
S:10			Active Nodes		
S:11			I/O Slot Enables		
S:12			I/O Slot Enables		
S:13			Math Register		
S:14			Math Register		
S:15			Node Address/ Baud Rate		
S:16			Debug Single Step Rung		
S:17			Debug Single Step File		
S:18			Debug Single Step Breakpoint Rung		
S:19			Debug Single Step Breakpoint File		
S:20			Debug Fault/ Powerdown Rung		
S:21			Debug Fault/ Powerdown File		
S:22			Maximum Observed Scan Time		
S:23			Average Scan Time		
S:24			Index Register		
S:25			I/O Interrupt Pending		
S:26			I/O Interrupt Pending		
S:27			I/O Interrupt Enabled		
S:28			I/O Interrupt Enabled		
S:29			User Fault Routine File Number		
S:30			STI Setpoint		
S:31			STI File Number		
S:32			I/O Interrupt Executing		
S:33			Extended Proc Status Control Word		
S:33/0			Incoming Command Pending		
S:33/1			Message Reply Pending		
S:33/2			Outgoing Message Command Pending		
S:33/3			Selection Status User/DF1		
S:33/4			Communicat Active		
S:33/5			Communicat Servicing Selection		
S:33/6			Message Servicing Selection Channel 0		
S:33/7			Message Servicing Selection Channel 1		
S:33/8			Interrupt Latency Control Flag		
S:33/9			Scan Toggle Flag		
S:33/10			Discrete Input Interrupt Reconfigur Flag		
S:33/11			Online Edit Status		
S:33/12			Online Edit Status		
S:33/13			Scan Time Timebase Selection		
S:33/14			DTR Control Bit		
S:33/15			DTR Force Bit		
S:34			Pass-thru Disabled		
S:34/0			Pass-Thru Disabled Flag		
S:34/1			DH+ Active Node Table Enable Flag		
S:34/2			Floating Point Math Flag Disable,Fl		
S:35			Last 1 ms Scan Time		
S:36			Extended Minor Error Bits		
S:36/8			DII Lost		
S:36/9			STI Lost		
S:36/10			Memory Module Data File Overwrite Protection		
S:37			Clock Calendar Year		
S:38			Clock Calendar Month		
S:39			Clock Calendar Day		
S:40			Clock Calendar Hours		
S:41			Clock Calendar Minutes		
S:42			Clock Calendar Seconds		
S:43			STI Interrupt Time		
S:44			I/O Event Interrupt Time		
S:45			DII Interrupt Time		
S:46			Discrete Input Interrupt- File Number		
S:47			Discrete Input Interrupt- Slot Number		
S:48			Discrete Input Interrupt- Bit Mask		
S:49			Discrete Input Interrupt- Compare Value		
S:50			Processor Catalog Number		
S:51			Discrete Input Interrupt- Return Number		
S:52			Discrete Input Interrupt- Accumulat		
S:53			Reserved/ Clock Calendar Day of the Week		
S:55			Last DII Scan Time		
S:56			Maximum Observed DII Scan Time		
S:57			Operating System Catalog Number		
S:58			Operating System Series		
S:59			Operating System FRN		
S:61			Processor Series		
S:62			Processor Revision		
S:63			User Program Type		
S:64			User Program Functional Index		
S:65			User RAM Size		
S:66			Flash EEPROM Size		
S:67			Channel 0 Active Nodes		
S:68			Channel 0 Active Nodes		
S:69			Channel 0 Active Nodes		
S:70			Channel 0 Active Nodes		

Address/Symbol Database

Address	Symbol	Scope	Description	Sym Group	Dev. Co
S:71			Channel 0 Active Nodes		
S:72			Channel 0 Active Nodes		
S:73			Channel 0 Active Nodes		
S:74			Channel 0 Active Nodes		
S:75			Channel 0 Active Nodes		
S:76			Channel 0 Active Nodes		
S:77			Channel 0 Active Nodes		
S:78			Channel 0 Active Nodes		
S:79			Channel 0 Active Nodes		
S:80			Channel 0 Active Nodes		
S:81			Channel 0 Active Nodes		
S:82			Channel 0 Active Nodes		
S:83			DH+ Active Nodes		
S:84			DH+ Active Nodes		
S:85			DH+ Active Nodes		
S:86			DH+ Active Nodes		
T4:0			Backwash Pending Trigger Delay		
T4:0/DN					
T4:1			High Pressure Alarm Delay		
T4:2			Low Flow Alarm Delay		
T4:3			Storage Tank LL Alarm Delay		
T4:4			Storage Tank HH Alarm Delay		
T4:5			Backwash Cycle Trigger Delay		
T4:6			Backwash Cycle		
T4:6.ACC					
T4:7			Fill Start Delay		
T4:8			Fill Stop Delay		
T4:9			Simulation Calculation Timer		
T4:10			Flow Switch Close Delay		
T4:10/DN					
T4:11			Backwash Cycle Count Reset Delay		
T4:12			Backwash Data Log Pulse Width		
T4:13			Hourmeter		
T4:13.ACC					
T4:14			Hourmeter Reset Delay		
U:3			IO		
U:4			Alarms		
U:5			Control Logic		
U:6			HOA		
U:7			Simulation		
U:8			System Mode		
U:9			Backwash Cycle		
U:10			Level Control		
U:11			Hourmeter		
U:12			Review / Testing Ladder		

Instruction Comment Database

Address	Instruction	Description

Symbol Group Database

Group_Name	Description